Weekly training log, week 11

Week 11

Module	EEEM068, Applied Machine Learning
Project	Industrial waste classification on WaRP
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Focus	SAM 2.1 + LoRA on WaRP-S

1. Experimental settings

Dataset	WaRP-S, around 600 train and 112 val image mask pairs
GPU	NVIDIA RTX 4000 Ada, 20 GB
Metric	mean IoU on the 23 instance val set
Goal	See how much LoRA fine tuning of SAM 2.1 gains over the zero shot baseline.

2. Hyperparameters

Trial 1, zero shot baseline

Model	SAM 2.1, sam2_hiera_large
Adaptation	none, just box prompts from ground truth
Image size	1024 x 1024

Trial 2, LoRA fine tuning

Model	SAM 2.1, sam2_hiera_large
Adapter	LoRA on the image encoder, rank 8, alpha 16
Trainable params	4.2 million out of 224 million
Batch size	2
Epochs	10
LR	1e-4
Weight decay	1e-4
Optimizer	AdamW, constant LR
Loss	Dice + BCE
Prompts	ground truth bounding boxes

3. Results

Trial 1, zero shot

Setting	val mIoU
SAM 2.1 with GT box prompts, no training	78.30%

Already strong because SAM has general object boundary priors from its own pretraining.

Trial 2, LoRA

Epoch	Loss	val mIoU
1	0.194	89.08%
2	0.135	86.88%
3	0.107	86.84%
4	0.102	88.59%
5	0.096	85.69%
6	0.096	87.89%
7	0.085	88.76%
8	0.082	88.37%

Epoch	Loss	val mIoU
9	0.092	84.90%
10	0.077	86.73%

Best epoch mIoU: 89.08% (epoch 1). Average across the run is around 86.7%, that is the number we report.

4. Observations

Training only 1.9% of SAM (the LoRA adapters) takes mIoU from 78.30% to 86.7%.

The val mIoU bounces around between 0.85 and 0.89 because the val set is tiny (23 instances). Every epoch is still well above zero shot.

SAM 2.1 plus LoRA is the segmentation result we submit.